Siyong Sung

M.S. Student, Mechanical and Aerospace Engineering | New Brunswick, NJ | syong.sung1@gmail.com | github.com/siyongsung | linkedin.com/in/siyong-sung-693a242b0 | siyongsung.github.io

SUMMARY

Controls-focused M.S. student developing wind-preview nonlinear MPC for gust rejection on multirotor UAVs, with the controller taken from simulation through PX4 software-in-the-loop to 47 indoor flights. Hands-on avionics background on F-16 aircraft (schematics, wiring, test equipment, technical documentation) and FAA A&P certified. Interested in flight control, GNC, and taking control algorithms from simulation to real hardware.

EDUCATION

Rutgers University — Master of Science, Mechanical and Aerospace Engineering 2025-01 – 2027-01 (expected)

New Brunswick, NJ · GPA 3.833 / 4.0

- Thesis: Wind Preview-Based Nonlinear Model Predictive Control for Gust Disturbance Rejection in Quadrotor UAVs (advisor: Prof. Laurent Burlion).
- Implemented the controller in Python / acados on ROS 2 and PX4, and validated it in MATLAB, PX4 software-in-the-loop, and 47 indoor flights against a fan wall under motion capture.

Southern Illinois University — Bachelor of Science, Aviation Technologies (Avionics) 2020-08 – 2024-12

Carbondale, IL · GPA 3.938 / 4.0

- Overhauled and functionally tested the complete avionics suite of a Cessna cockpit on a test bench (senior project).
- Designed and built an aircraft landing-gear warning system on Arduino.

Orange Coast College — Associate of Science, Aviation Maintenance Technology 2018-08 – 2020-08

Costa Mesa, CA · GPA 3.38 / 4.0

- FAA A&P curriculum: airframe and powerplant maintenance, inspection, troubleshooting, and regulations; six-cylinder engine overhaul to the manufacturer's manual.
- Avionics Certificate of Achievement (wiring practices, basic avionics troubleshooting).

PROFESSIONAL EXPERIENCE

Republic of Korea Air Force, 19th Fighter Wing — Avionics Technician (Technical Sergeant) 2021-09 – 2023-12

Chungju, South Korea

- Scheduled inspections, troubleshooting, and corrective maintenance on F-16 and KF-16 avionics systems using technical orders, schematics, and test equipment.
- Documented maintenance actions, discrepancies, and parts requests to squadron procedures.
- Supported a Lockheed Martin field engineer with Korean–English technical interpretation during an investigation of an F-16 communication system.

LICENSES & CERTIFICATIONS

- FAA Airframe & Powerplant (A&P) License — Federal Aviation Administration, 2020-12
- FCC General Radiotelephone Operator License (GROL) — Federal Communications Commission, 2020-03
- NCATT Aircraft Electronics Technician (AET) — National Center for Aerospace & Transportation Technologies, 2019-03

SKILLS

- **Controls:** Model predictive control, flight dynamics and control, system identification, flight testing
- **Software:** MATLAB / Simulink, Python, ROS 2, PX4, Gazebo, acados
- **Avionics and hardware:** Avionics diagnostics, wiring and schematics, test equipment, technical documentation

HONORS & AWARDS

- High Achievers Alternate Tuition Rate (scholarship) — Southern Illinois University
- Dean's List — Southern Illinois University
- Dean's List — Orange Coast College

LANGUAGES

- Korean (Native)
- English (Fluent)